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BIT 2223 Mobile Computing

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 1

Overview of technologies for wireless communication

Objectives

Upon completing this topic, you should be able to:

- *Understand the concept of Mobile computing basics*
- *Explain some of the wireless data transmission methods*
- *Understand the benefits of wireless networks*
- *Understand the concept of data packet transfers*
- *Understand the concept of trunking*





1.1. Introduction

Mobile computing is a term describing one's ability to use technology while moving, as opposed to portable computers, which are only practical for use while deployed in a stationary configuration. Mobile computing runs on stand-alone applications (or access remote applications) via: – wireless networks: IR, Bluetooth, W-LANs, Cellular, W-Packet Data networks, SAT. etc.

There are several forms of computing:

- Wireless Computing
- Nomadic Computing
- Mobile Computing
- Ubiquitous Computing
- Pervasive Computing



- Invisible Computing
- Metamorphic Computing

1.2. Wireless vs. Mobile Communications

Wireless telecommunications can be divided into two broad categories: mobile communications and fixed wireless communications. Each category has its own unique market in terms of customer needs and technology requirements. The mobile communications market requires mobility or non-tethered communications.

The goal of mobility is anytime, anywhere communications. Mobile communications technology must be able to allow roaming - the ability to provide service to mobile phone users while outside their home system. On the other hand, fixed wireless is



BIT 2223 Mobile Computing

simply an alternative to wired communications. The fixed wireless user does not need mobility. Instead, the fixed wireless user needs cost effective telecommunications from fixed locations. Wireless is an alternative means of providing service. It is sometimes the only means. When the customer is in a remote location, satellite is the only alternative.

1.3. Cellular and PCS

The personal communications concept arose after cellular networks were deployed. Personal Communications Service (PCS) technologies were designed to meet the needs of anytime, anywhere personalized communications. PCS networks were deployed utilizing cellular RF designs similar to cellular. However, many PCS carriers initially deployed larger groupings of smaller





BIT 2223 Mobile Computing

cell sites to cover densely populated urban areas. PCS also uses a higher portion of the RF spectrum (1900 MHz in the US versus 800 MHz for cellular).

Being deployed after the initial cellular networks, PCS networks also initially had more advanced technologies than PCS, including SS7 network infrastructure for services such as calling number identification. However, cellular would soon catch up due to competitive pressures. In aggregate, there are now no substantive differences between the initial "cellular" networks and "PCS". In fact, they both utilize the same underlying technologies.



1.4. Limitations of mobile computing

- **Insufficient bandwidth:** Mobile internet access is generally slower than direct cable connections, using technologies such as GPRS and EDGE, and more recently 3G networks. These networks are usually available within a range of commercial cell phone towers. Higher speed wireless LANs are inexpensive, but have very limited range.
- **Security standards:** When working, mobile is dependent on public networks, requiring careful use of VPNs.
- **Power consumption:** When a power outlet or portable generator is not available, mobile computers must rely entirely on battery power. Combined with the compact size of many mobile devices, this often means unusually expensive



sive batteries must be used to obtain the necessary battery life.

- **Transmission interferences:** Weather, terrain, and the range from the nearest signal point can all interfere with signal reception. Reception in tunnels, some buildings, and rural areas is often poor.
- **Potential health hazards:** More car accidents are related to drivers who were talking through a mobile device. Cell phones may interfere with sensitive medical devices. There are allegations that cell phone signals may cause health problems.



1.5. Wireless Network

Wireless network refers to any type of computer network that is implemented without the use of wires. Wireless telecommunications networks are generally implemented with some type of remote information transmission system that uses electromagnetic waves, such as radio waves, for the carrier and this implementation usually takes place at the physical layer of the OSI model.

Wire or fiber-based data communications spans a huge range of throughputs and distances—28.8 Kbps over a modem connection; 10/100 Mbps over an Ethernet segment; and gigabit speeds over fiber. Similarly, wireless connections span a huge range. The world of wireless data includes:

- fixed microwave links,



- wireless LANs,
- data over cellular networks,
- wireless WANs,
- satellite links,
- digital dispatch networks,
- one-way and two-way paging networks,
- Infrared, laser-based communications, keyless car entry,
- The Global Positioning System and more

1.6. Application of Mobile Computing

Most computer services are moving from E to M service e.g M-Banking, M-Commerce, M-payment etc. The different application areas of mobile computing are :



1. Vehicles

- Transmission of news, road condition etc: It will be useful for the passengers to know this information. This helps them to choose better path towards destination.
- Ad-hoc network near vehicles to prevent accidents

2. Emergencies

- Portability helps a lot in early transmission of patient data to the hospital.
- Ad-hoc network in case of earthquakes: cyclones are very useful since usual modes of communication like telephone etc will not be in danger of falling apart.
- Military applications include communication between soldiers in the battleground.



3. Traveling salesmen

- Direct access to central customer files
- Consistent databases for all agents
- Availability of a Mobile office

4. Web access

- Outdoor Internet access
- Intelligent travel guide with up-to-date location dependent information

5. Information services

- Push: stock quotes;
- Pull: nearest cash ATM
- Nearest printer

6. Disconnected operations



- File-system caching for offline work
- Mobile agents, e.g., shopping

7. Entertainment

- Ad-hoc networks for multi user games
- VOD (video on Demand)

1.7. Limitations of Mobile Environments

There are some specific limitations of the mobile environments which can be classified as :

1. Limitations of the wireless network

- **Heterogeneous Network:** The different wireless networks might be heterogeneous in nature in the sense that different networks might follow different



standards and hence communication between them can become quite difficult.(CDMA and GSM more expensive)

- **Frequent disconnection:** Since wireless network uses mostly space as the medium, the medium itself becomes quite unreliable and can lead to frequent problems of disconnection. (weather conditions)
- **Insufficient bandwidth:** Mobile internet access is generally slower than direct cable connections, using technologies such as GPRS and EDGE, and more recently 3G networks. These networks are usually available within range of commercial cell phone towers. Higher speed wireless LANs are inexpensive, but have very limited range.



2. Limitations imposed by mobility

The mobility of the nodes in a network can place additional constraints especially on the applications running on these nodes, which otherwise is transparent to the mobility of its host. Due to the mobility of the hosts, it can happen that the route through which two nodes were communicating gets broken. During hand-off, the mobile experiences slight disconnection.

- Lack of mobility-awareness by the system
 - *network*: existing transport protocols are inefficient to use across heterogeneous mix of fixed/wireless networks
 - *session and presentation*: inappropriate for the wireless environment and for mobility



- *operating systems*: lack of environmental related conditions and signals
- *client/server*: unless changed they are inappropriate and inefficient

3. Limitations of the mobile computer

- Power: One of the main limitations imposed by the mobile node is its battery power (max~ 5 hours). The mobile nodes use battery for their operations, so the battery power is a valuable resource which must be conserved at all cost.
- Device limitations (limited capability):
 - E.g., small LCD on a mobile telephone can only display a few lines of text
 - E.g., browsers of most mobile wireless devices



use wireless markup language (WML) instead of HTML which is the defacto standard.

- E.g., memory, input devices, and disk space
- To make the nodes mobile, we need to make the devices as small and portable as possible. This will require the sacrifice of some computing or storage capacities of the device.
- Very unreliable: Subject to theft and destruction
- Highly unavailable (normally powered-off to conserve battery)

4. Limitations of the Communication Bandwidth

- Orders of magnitude slower than fixed network
- Higher transmission bit error rates (BER)
- Uncontrolled cell population



- Difficult to ensure Quality of Service (QoS)
- Asymmetric duplex bandwidth
- Limited communication bandwidth exacerbates the limitation of battery lifetime.

1.8. Frequent Mobile Computing Disconnections

- Handoff blank out ($>1\text{ms}$ for most cellulars)
- Drained battery disconnection
- Battery recharge down time
- Voluntary disconnection (turned off to preserve battery power, also off overnight)
- Theft and damage (hostile environment)
- Roam-off disconnections



1.9. Benefits of wireless networks

- connections when none are possible,
- connections at lower cost in many scenarios-no costing of cabling
- faster connection
- Portability, mobility, flexibility
- wireless networks are much faster to install



Revision Questions

EXERCISE 1.  List 4 application areas of mobile computing

Example . Explain implementation of wireless network

Solution: Wireless networks are implemented by use remote information transmission systems that uses electromagnetic waves such as radio waves. ☐

EXERCISE 2.  Explain the limitations of mobile computing



Solutions to Exercises

Exercise 1.

Vehicles, Emergencies, Web Access & Information Services.

Exercise 1



Exercise 2.

Insufficient bandwidth: Mobile internet access is generally slower than direct cable connections, using technologies such as GPRS and EDGE, and more recently 3G networks.

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BIT 2223 Mobile Computing

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Exercise 2